Exercise 6: Rule-Based Learning in 2D

(a) Formulate the most general hypothesis h_{g0}

The most general hypothesis must cover every possible point in the feature space. Since both coordinates range from 1 to 10, the rectangle must cover the entire region.

$$\mathbf{h_{g0} = (1, 1, 10, 10)}$$

This rectangle includes all possible examples, from (1, 1) to (10, 10).

(b) Check the more-general relation $\geq_g$

A hypothesis h1 = (a, b, c, d) is more general than h2 = (e, f, g, h) if the rectangle of h1 contains the entire rectangle of h2. The required conditions are:

$$\mathbf{a \leq e, \quad b \leq f, \quad c \geq g, \quad d \geq h}$$

Statement	Check	Result
(1, 2, 3, 4) ≥g (1, 1, 4, 4)	2 ≤ 1 is false; 3 ≥ 4 is false.	False
(2, 3, 6, 7) ≥g (3, 4, 5, 7)	2 ≤ 3, 3 ≤ 4, 6 ≥ 5, 7 ≥ 7: all true.	True
(1, 1, 2, 8) ≥g (1, 1, 3, 3)	2 ≥ 3 is false.	False
(3, 3, 9, 9) ≥g (1, 1, 1, 1)	3 ≤ 1 is false (for both lower bounds).	False

Final answer (b): Only the second statement is correct.

(c) Determine two minimal specializations

Given h = (2, 3, 5, 7) and the negative example x = (2, 7), c = 0. The point (2, 7) lies inside the original rectangle because 2 ≤ 2 ≤ 5 and 3 ≤ 7 ≤ 7. We must shrink the rectangle so that it excludes this negative point.

First minimal specialization: increase the lower x-coordinate from 2 to 3. This excludes the point because its x-coordinate is 2.

$$\mathbf{h1 = (3, 3, 5, 7)}$$

Second minimal specialization: decrease the upper y-coordinate from 7 to 6. This excludes the point because its y-coordinate is 7.

$$\mathbf{h2 = (2, 3, 5, 6)}$$

Both rectangles are subsets of the original rectangle and exclude the negative example. Neither can be made more specific while remaining a minimal specialization.

Final answers summary

Part	Answer
(a)	h_{g0} = (1, 1, 10, 10)
(b)	Only statement 2 is true.
(c)	h1 = (3, 3, 5, 7) and h2 = (2, 3, 5, 6)